

Public Beach Water

Consistent with O. Reg. 671/92 of the *French Language Services Act*, laboratory testing information on this page is only available in English because it is scientific or technical in nature and is for use only by qualified health care providers and not by members of the public.

Background

This page provides public beach water testing information for recreational water submitted to the Public Health Ontario (PHO) laboratory by an official agency. Official agency clients include public health units, Ministry of Natural Resources and Forestry Ontario Parks, and other agencies when authorized for testing.

The target organism of the test is *Escherichia coli* (*E. coli*). The following testing options are available for recreational water: Membrane filtration for *E. coli*.

For information regarding other water testing options, refer to the [Test Information Index](#). For general inquiries related to water sample collection, submission and testing, please contact PHO's laboratory [Customer Service Centre](#).

Updates

- Added Submitter's Responsibility section
- Updated the following sections: Submission and Collection Notes (use aseptic technique and avoid getting sediment in bottle), Storage and Transport (store samples in the dark and prevent direct contact with ice packs) and Testing Methods (added assay performance and limitations)

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Testing Indications

[Guidelines\ for Canadian Recreational Water Quality](#)¹

[Recreational Water Protocol](#)²

[Operational Approaches for Recreational Water Guideline](#)³

Acceptance/Rejection Criteria

Refer to the [Public Health Inspector's Guide to Environmental Microbiology Laboratory Testing](#).⁴

Submitter's Responsibility

Refer to the Ministry of Health [Recreational Water Protocol](#)² and [Operational Approaches for Recreational Water Guideline](#).³

By submitting the water sample for testing, the submitter accepts Public Health Ontario's methodology, and represents and warrants that the water sample was taken from the Location of Water Source indicated in the test requisition and that the information provided is true in all material respects at the time of submission. The Ontario Agency for Health Protection and Promotion assumes no responsibility for the accuracy of the information provided, the manner in which the sample was collected or the mode by which it was transported to the laboratory.

Prior to the beginning of each public beach water monitoring season, notify your local **PHO laboratory** of the number of samples that will be submitted, day(s) of week for sampling (Note: routine samples are only accepted at the laboratory Monday – Friday), and approximate time of day that samples will arrive at the laboratory.

For legal samples, the chain of custody/requisition must be complete and accurate. To maintain the chain of custody of the sample, the “Relinquished by” section must be completed **AND** a Legal seal with all the required information applied over the cap when the sample is received at the laboratory. Refer to **Instructions For Official Agencies Submitting Water Samples to the Public Health Ontario Laboratory**.⁵

Specimen Collection and Handling

Specimen Requirements

Test Requested	Required Requisition(s)	Specimen Type	Minimum Volume	Collection Kit
<i>E. coli</i>	Bacteriological Analysis of Water: Multiple Sample Requisition for Official Agencies	Public Beach Water	200 mL (to the fill line on the bottle)	Sterile Water bottles - 250 mL capacity (Case of 250 bottles) Item #: 300013

Submission and Collection Notes

1

Complete the following fields of the requisition:

- “Official Agency Address” – include the sub office if this office submitted the sample
- “Sample Information - Non-Potable” section
- “Identification of Collection Site & Time Collected” section
- “For Regulated Drinking Water or Legal Samples” section – as applicable (refer to **Instructions For Official Agencies Submitting Water Samples to the Public Health Ontario Laboratory**)⁵

Note: The sample may not be tested if the required information is incomplete and/or inaccurate when the sample is received at the laboratory.

2

Remove one barcode from the bottle and apply it to the top copy of the requisition in the “Barcode” field. Remove a second barcode and apply it in the corresponding location on the second copy of the form. Retain a copy of the completed requisition that includes the barcodes.

Note: Samples may not be processed if the barcodes are not affixed to the requisition. If the barcode is not able to be removed, write the barcode number in the “Barcode” box on the form.

3

Examine the lid of the bottle. If the tamper evident ring has separated from the cap use another bottle.

4

Wear clean, disposable gloves or have clean, sanitized hands prior to collection and keep the bottle closed until just before collection. Remove cap and ensure it is kept away from sources of contamination.

5

Do not touch the bottle mouth and neck as well as inside of the lid or bottle with hands or other surfaces, do not put the lid down, and do not ingest the sodium thiosulfate in the bottle. For accidental exposure get medical attention.

6

Collect a minimum of five (5) samples where possible at designated sites at each identified beach location. Sampling locations should reflect water quality within the entire recreational zone. More than five (5) samples may be required for larger beaches. Note: One requisition can accommodate six (6) samples.

7

Follow the instructions in the **Operational Approaches for Recreational Water Guideline**.³

Alternatively, plunge bottle downward into water to 30 cm – turn bottle in direction point of current so the bottle mouth is pointed on the upstream side of the sampler (i.e., away from the sampler). When sampling in the surf zone, samples should be collected during incoming waves or surges of water. Select areas of greatest bather load and note the following.⁶

- a. Samples may be taken ankle deep (at approx. 7.5 cm below water surface).
- b. In deeper waters, if desired, take another sample approximately 7.5 cm below the water surface. This area may be somewhere between the knees and the chest, depending on how deep the water is where the sample is taken.
- c. Take samples over the range of environmental and climatic conditions, especially during times when maximal pollution can be expected (i.e., periods of tidal, current, and wind influences; storm water runoff; and wastewater treatment bypasses).
- d. Avoid contamination from floating debris and minimize the collection of suspended sands and sediments in the sample.

8

Collect the sample to the 200mL fill line. Do not overfill the bottle. If overfilled, remove some of the water so the bottle is just filled to 200 mL line.

9

Tighten the cap on the collection bottle. Leaking samples will not be processed; a new sample and requisition will be required.

10

Refer to **Instructions for Official Agencies Submitting Water Samples to the Public Health Ontario Laboratory**⁵ for step by step instructions for completing the documentation and submitting/relinquishing the sample.

Timing of Specimen Collection

Ensure adequate time to collect and transport the sample to the laboratory. All public beach water samples must be shipped within 24 hours of collection and tested within one calendar day of collection.

Routine Public Beach water samples are accepted Monday to Friday 8:30 a.m. to 4:30 p.m. excluding statutory holidays. If the statutory holiday falls on a weekday, samples will not be accepted the day before the holiday as well.

Please contact PHO's laboratory **Customer Service Centre** prior to the submission of samples that will be received outside of regular operating hours (Monday to Friday 8:30 a.m. to 4:30 p.m. excluding statutory holidays). Samples of an urgent nature (e.g., STAT will be processed and read with no undue delay).

Limitations

Samples are only accepted when submitted by a public health unit, MNRF Ontario Park or other agencies when authorized for testing.

Samples may not be tested if the acceptance criteria are not met. Refer to the [Public Health Inspector's Guide to Environmental Microbiology Laboratory Testing](#)⁴ for details.

Due to unforeseen circumstances, it may be necessary to refer samples to another laboratory for testing, other than the laboratory to which sample was initially submitted.

Storage and Transport

Keep specimens in the dark, stored at 2 – 8 °C following collection and ship them to **PHO's laboratory** inside insulated containers or in a cooler with frozen ice packs (Note: Do not include medical samples in the cooler). Arrange the samples so they do not tip and avoid direct contact with ice packs. Frozen samples are not accepted.

Note: Samples must be shipped to the local **PHO Laboratory** within 24 hours of collection and tested within one day of collection.

Special Instructions

It may be necessary to refer samples to another laboratory for testing, other than the laboratory to which sample was initially submitted.

Refer to the Ministry of Health [Recreational Water Protocol](#)² and [Operational Approaches for Recreational Water Guideline](#)³ for information related to recreational water protocols.

To order collection kits or other PHO laboratory supplies, complete the [Requisition for Specimen Containers and Supplies](#). Fax the form to the PHO Toronto laboratory fax number listed on the [Requisition for Specimen Containers and Supplies](#) or your local **PHO laboratory**.

Requisitions and Kit Ordering

TEST REQUISITION



Bacteriological Analysis of Water: Multiple Sample Requisition for Official Agencies

Laboratory Test Requisition Form. This form addresses the bacteriological analysis of water for Official Agency submitter.

77 KB | Updated 16 May 2024

Test Frequency and Turnaround Time (TAT)

Samples are routinely accepted and tested at the laboratory Monday – Friday during regular operating hours 8:30 a.m. to 4:30 p.m. For submission around statutory holidays: Refer to the PHO laboratory holiday schedule. Please contact PHO's laboratory **Customer Service Centre** prior to the submission of samples that will be received outside of these hours.

Turnaround time (TAT) is up to 4 business days from receipt at **PHO's laboratory**.

STAT and Critical Samples Testing

Please contact PHO's laboratory **Customer Service Centre** prior to the submission of urgent samples or those that will be received outside of regular operating hours. If outside of Customer Service's hours, contact the PHO laboratory Duty Officer at (416) 605-3113.

STAT samples must be identified with "STAT" on the requisition. STAT samples and samples submitted under a drinking water regulation will be processed and read with no undue delay.

Test Methods

The public beach water *E. coli* test is performed by the Membrane Filtration method modified from the American Society for Microbiology's Bacteriological Examination of Waters: Membrane Filtration Protocol.⁷

Assay performance and limitations:

E. coli assay performance:

- In-house studies show that more than 90% of *E. coli* strains present in water can be detected using this procedure, however, false positives and false negatives can occur. Three percent (3%) of *E. coli* isolates are β -glucuronide – negative.⁸ This method does not allow for the determination of toxigenic species of *E. coli* (e.g., *E. coli* 0157:H7)

Membrane filtration assay limitations:

- Material that is larger than the pore size of the membrane filter (e.g., particulate matter or algae) can clog the membrane filter and interfere with target colony detection
- Bacterial cells that have been stressed, e.g., exposed to adverse environmental conditions, may not survive the filtration process or may be negatively affected by the chemicals used as selective agents in the medium.
- Some non-target bacteria may release bacteriocins, proteins that can inhibit the growth of target bacteria. High levels of toxic metals or organic compounds may become concentrated in the membrane filter and inhibit colony growth

Interpretation

Refer to the **Public Health Inspector's Guide to Environmental Microbiology Laboratory Testing**.⁴

Reporting

Results are reported to the submitter as indicated on the requisition and as per PHO laboratory's reporting protocol.

References

1. Health Canada. Guidelines for Canadian recreational water quality. 3rd ed. Ottawa, ON: Her Majesty the Queen in Right of Canada, represented by the Minister of Health; 2012 [cited 2023 Sep 21]. Available from: <https://www.canada.ca/en/health-canada/services/publications/healthy-living/guidelines-canadian-recreational-water-quality-third-edition.html>
2. Ontario. Ministry of Health. Recreational water protocol, 2019 [Internet]. Toronto, ON: King's Printer for Ontario; 2019 [cited 2023 Sep 21]. Available from: <https://www.ontario.ca/page/ontario-public-health-standards-requirements-programs-services-and-accountability> (Access under the Protocols section of the Ontario Public Health Standards)
3. Ontario. Ministry of Health. Operational approaches for recreational water guideline, 2018 [Internet]. Toronto, ON: King's Printer for Ontario; 2018 [cited 2023 Sep 21]. Available from: <https://www.ontario.ca/page/ontario-public-health-standards-requirements-programs-services-and-accountability> (Access under the Guidelines section of the Ontario Public Health Standards)
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5. Ontario Agency for Health Protection and Promotion (Public Health Ontario). Instructions for official agencies submitting water samples to the Public Health Ontario laboratory [Internet]. Toronto, ON: Queen's Printer for Ontario; 2012 [cited 2023

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6. Lipps WC, Braun-Howland EB, Baxter TE, editors. Standard Methods for the Examination of Water and Wastewater. 24th ed. Washington, DC: APHA Press; 2023.
7. Arango Pinedo C, Foster B. Bacteriological examination of waters: membrane filtration protocol [Internet]. Houston, TX: American Society for Microbiology; 2015 [cited 2023 Sep 21]. Available from: <https://asm.org/ASM/media/Protocol-Images/Bacteriological-Examination-of-Waters-Membrane-Filtration-Protocol.pdf?ext=.pdf>
8. Kilian M, Bülow P. Rapid identification of Enterobacteriaceae. II. Use of a β -glucuronidase detecting agar medium (PGUA agar) for the identification of *E. coli* in primary cultures of urine samples. Acta Pathol Microbiol Scand B. 1979;87(5):271-6. Available from: <https://pubmed.ncbi.nlm.nih.gov/393074/>

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Contact Laboratory Customer Service

Microbiology and Laboratory Services

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